

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – SHORT ANSWER TEST (30 Questions)

Course Code:	DSA06	Course Title:	Data Handling and Visualization
CO Assessed:	CO4 – Naturalization (BL5)	Assessment:	Assessment Tool 1 – Short Answer Test
Weightage:	40% of CIA	Total Marks:	30 (6 Questions × 5 Marks)
CO4 Statement:	<i>Evaluate visualization designs based on perception, cognition, and effectiveness principles (BL5).</i>		
Student Name:	G Kushwanth	Reg. No.:	192424150
Section / Batch:	AI&DS	Date:	

Code of Conduct

I G Kushwanth(192424150) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: G Kushwanth

Instructions

- This test contains 6 questions, each carrying 5 marks. Total: 30 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic	Focus	Q Nos	Marks
Scatterplots & Correlation Analysis	Correlation, outliers, and relationship identification using scatterplots	1	5
Correlograms & Feature Selection	Correlation matrices and feature importance analysis	2	5
Dimension Reduction (PCA, t-SNE, UMAP)	High-dimensional data visualization and comparison of techniques	3, 4	10

Bubble Charts & Multivariate Visualization	Encoding multiple variables and interpreting limitations	5	5
Hexbin Plots & Dense Data Visualization	Handling overplotting in large datasets	6	5
GRAND TOTAL:		30 Marks	

Annexure A – Short Answer Test

1. How does a violin plot provide more information than a box plot when analyzing data distribution?

a) What does the width of a violin plot represent?

b) How can a violin plot help identify multimodal distributions? Answer:

A violin plot overlays a mirrored kernel density estimate (KDE) on top of the box-plot summary (median, quartiles, whiskers), so it shows the complete shape of the distribution rather than only five summary statistics.

a) The width of the violin at any given value represents the estimated density of data points at that value — a wider section means more observations are concentrated there, and a narrower section means fewer.

b) Multimodal distributions appear as multiple distinct bulges or peaks along the violin's shape; a standard box plot cannot reveal this because it collapses the data into a single median and IQR, hiding the presence of more than one cluster/mode.

2. How does a pair plot help in exploring relationships among multiple variables in a dataset?

a) What information is displayed along the diagonal of a pair plot? b) How can pair plots assist in detecting clusters and correlations? Answer:

a) The diagonal cells display the univariate distribution of each individual variable — typically as a histogram or a KDE curve — since plotting a variable against itself would just be a straight line. **b)**

Clusters show up as groups of points that separate spatially within an off-diagonal panel (often highlighted further by colouring points by class/category), while correlations appear as a visible linear or curved trend between two variables across their panel.

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3. What is the purpose of a heatmap in data visualization and

analysis? a) How are values represented in a heatmap?

b) Why are heatmaps useful for identifying patterns and anomalies? Answer:

A heatmap uses colour intensity/hue instead of numbers to represent the magnitude of values in a matrix (such as a correlation matrix or a table of measurements), making large tables of numbers easy to scan visually.

a) Each cell's value is mapped onto a colour scale (e.g., a colour gradient from blue for low/negative values to red for high/positive values), so the colour itself communicates the value's magnitude and sign.

b) Because similar values map to similar colours, heatmaps let the eye quickly pick out blocks of correlated variables or repeated patterns, while any cell that breaks the surrounding colour pattern stands out immediately as a potential anomaly or outlier.

4. How does a parallel coordinates plot visualize high-dimensional

data? a) What does each vertical axis represent?

b) How can this plot be used to compare observations across multiple variables? Answer:

A parallel coordinates plot visualizes high-dimensional data by placing one vertical axis for each variable side by side, then drawing a connected polyline for every observation that passes through its value on each axis in turn.

a) Each vertical axis represents one variable/feature in the dataset, independently scaled to that variable's own range of values.

b) Comparing the slope and path of lines between adjacent axes reveals relationships — near-parallel lines between two axes indicate a positive correlation, while many crossing lines indicate a negative correlation — and tracing or highlighting a single observation's polyline across all axes shows how that one record compares with others on every variable simultaneously.

5. How does a contour plot represent three-dimensional data on a two-dimensional surface?

a) What do contour lines indicate?

b) In which applications are contour plots commonly used?

Answer:

A contour plot represents three-dimensional data (two continuous input variables plus a response/height value) on a flat two-dimensional surface by drawing curves that connect all points sharing the same value of the third (z) variable, similar to a topographic map.

a) Each contour line marks a constant value of the third variable; closely spaced contour lines indicate a steep gradient (rapid change), while widely spaced lines indicate a gentle slope, and closed loops mark local maxima or minima.

b) Contour plots are commonly used in weather maps (pressure/temperature fields), topographic and geographic maps (elevation), engineering and physics (stress/field visualization), and in machine learning to visualize loss/objective-function surfaces during optimization.

6. How does a density plot help in understanding the distribution of continuous data?

a) How is a density plot different from a histogram?

b) What does a peak in a density plot indicate about the data?

Answer:

A density plot estimates and displays the underlying probability distribution of continuous data as a smooth, continuous curve, usually computed using kernel density estimation (KDE). **a)** Unlike a histogram, which groups data into discrete bins whose appearance depends heavily on bin width and boundary choice, a density plot produces a smooth continuous curve that avoids this binning artefact and gives a cleaner estimate of the distribution's shape.

b) A peak in a density plot indicates a mode — a value or range where data points are most concentrated; a single peak indicates a unimodal distribution, while multiple peaks indicate a multimodal distribution, and the peak's height reflects the relative density of data at that point.